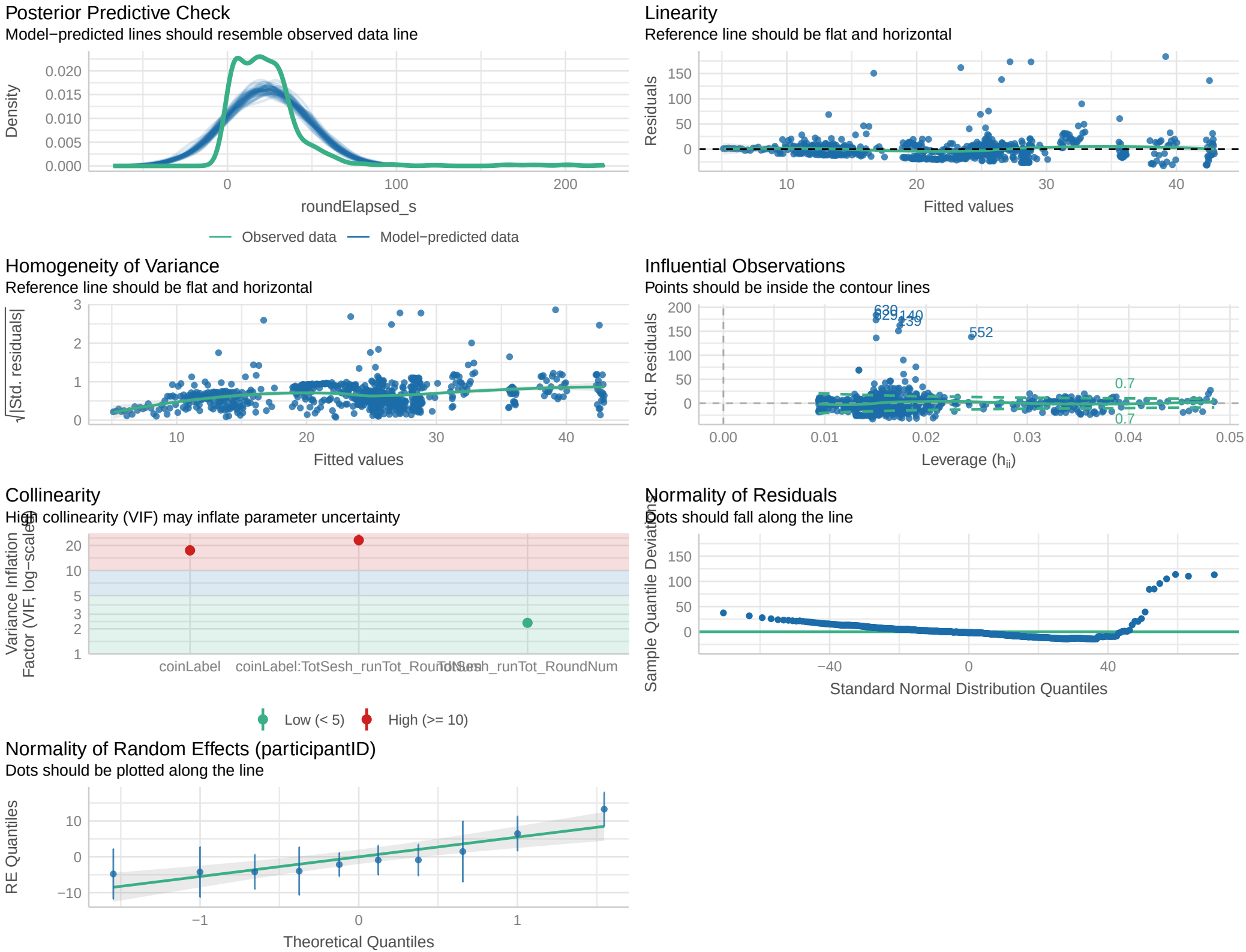
Posterior Predictive Check
Model−predicted lines should resemble observed data line
Density
0.020
0.015
0.010
0.005
0.000
0
100
200
roundElapsed_s
Observed data
Model−predicted data
Linearity
Reference line should be flat and horizontal
Residuals
150
100
50
0
10
20
30
40
Fitted values
Homogeneity of Variance
Reference line should be flat and horizontal
√|Std. residuals|
3
2
1
0
10
20
30
40
Fitted values
Influential Observations
Points should be inside the contour lines
Std. Residuals
200
150
100
50
0
630
629
140
139
552
0.7
0.7
0.00
0.01
0.02
0.03
0.04
0.05
Leverage (hii)
Collinearity
High collinearity (VIF) may inflate parameter uncertainty
Variance Inflation Factor (VIF, log−scale)
20
10
5
3
2
1
coinLabel
coinLabel:TotSesh_runTot_RoundNum
TotSesh_runTot_RoundNum
Low (< 5)
High (>= 10)
Normality of Residuals
Dots should fall along the line
Sample Quantile Deviations
150
100
50
0
−40
0
40
Standard Normal Distribution Quantiles
Normality of Random Effects (participantID)
Dots should be plotted along the line
RE Quantiles
10
0
−10
−1
0
1
Theoretical Quantiles